The Local Whitehead-Move Chain on Cubic Multigraphs

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Abstract

We study the random walk on the graph of connected cubic multigraphs under Whitehead moves. For even $N \geq 2$, the states are isomorphism classes of graphs with N vertices. At each step, an edge and one of the two Whitehead moves on it are chosen uniformly. Loops and multiple edges are allowed, and both moves on a loop act trivially. Moves with isomorphic results are counted with multiplicity.

We prove that the chain is reversible and determine its stationary distribution: the cubic configuration model conditioned on connectedness, viewed on isomorphism classes. For its transition matrix P_N, we show that $\mathrm{Spec}(P_N) \subseteq [-1/2, 1]$. Functions of the number of loops give, for every fixed r, the bound $2r/(3N) + o(N^{-1})$ on the $(r+1)$st smallest eigenvalue of $I - P_N$. We also prove that the cut between bridged and bridgeless graphs has conductance exactly $1/(3N/2 - 1)$. These estimates give linear lower bounds for relaxation and mixing times.

An exhaustive enumeration gives the full transition matrices for every even N from 2 through 14, reaching 21,096 classes and 886,032 Whitehead moves at $N = 14$. The computations verify the stationary weights, bridge flow, and exact conductance formula.

Keywords: Whitehead move, cubic multigraph, configuration model, Markov chain, conductance, reconfiguration graph.
2020 Mathematics Subject Classification: 05C81, 05C40, 60J10, 05C30.

1 Introduction

The graph of cubic graphs has isomorphism classes of connected cubic multigraphs as vertices and Whitehead moves as directed edges. Such a move collapses a non-loop edge and expands the resulting four-valent vertex in one of two other ways. We allow loops and multiple edges, and a move on a loop is defined to leave the graph unchanged. There is one directed edge for each move, even when several moves have isomorphic results.

Whitehead moves also arise from pants decompositions. When $N = 2g - 2$, a cubic multigraph can be used as the dual graph of a pants decomposition of a closed surface of genus g: put one vertex in each pair of pants and one edge across each glued cuff. The surface retains gluing data that the abstract graph forgets. Hatcher describes the corresponding complex of pants decompositions [4]. Rafi and Tao determine the asymptotic diameter for single and simultaneous Whitehead moves and relate these metrics to the geometry of moduli space [11].

The expansion question raised by Rafi was answered by Grave de Peralta and Kolpakov [3]: they used the cut between bridged and bridgeless graphs to prove that outer conductance tends to

zero. Their outgoing-edge volume assigns the same weight to every class, since each class has $3N$ outgoing moves. Li subsequently proved that increasingly many eigenvalues of the simple labelled and unlabelled graph Laplacians remain bounded [7].

Here we consider the random walk that chooses an edge of the cubic graph and one of its two Whitehead moves uniformly. Its transition matrix is obtained by dividing the directed adjacency matrix by $3N$. Although each move is reversible, the numbers of moves between two isomorphism classes need not agree in the two directions. In particular, the uniform distribution on classes need not be stationary.

Our results determine the stationary law and give explicit spectral and conductance bounds for this multiplicity chain. The proof lifts the walk to labelled half-edge configurations, where each move is a symmetric exchange. Conditioning on the chosen edge gives a Johnson-graph operator on $J(4,2)$; counting loop removals gives an exact Dirichlet form and Charlier-polynomial test spaces. We also give a cycle-based proof of the bridge-crossing rule from [3] and compute the stationary flow across that cut.

More precisely, let m_{ii} count the loops at vertex i, let m_{ij} count the edges between distinct vertices i and j, and let $\mathrm{Aut}_v(G)$ be the group of vertex permutations preserving these multiplicities. The stationary weight of the class of G is proportional to

$$\frac{1}{|\operatorname{Aut}_v(G)|}\,\frac{(3!)^N}{\prod_i 2^{m_{ii}} m_{ii}! \prod_{i<j} m_{ij}!}. \tag{1.1}$$

We prove that $\mathrm{Spec}(P_N) \subseteq [-1/2, 1]$ and that, for every fixed r,

$$\lambda_{r+1}(I - P_N) \le \frac{2r}{3N} + o(N^{-1}).$$

For the bridged–bridgeless cut we obtain conductance $1/(3N/2-1)$ for every even $N \ge 2$. The proof uses the formal generating functions of connected and bridgeless cubic graphs, related by the combinatorial Legendre transform [2, Section 5.8]. Hence, for fixed $\varepsilon < 1 - e^{-1}$,

$$t_{\mathrm{mix}}(\varepsilon) \ge \frac{3N}{2}\big(1 - e^{-1} - \varepsilon\big)(1 + o(1)).$$

The stationary distribution is given in Theorem 3.3. The spectral and loop-count estimates are proved in Proposition 5.1 and Theorem 6.2, and the conductance and mixing bounds in Theorems 7.2 and 7.3 and Corollary 7.4. Finally, Section 8 gives an exhaustive enumeration for $N = 2, 4, 6, 8, 10, 12, 14$. All limits are taken through even N.

2 The Local Whitehead Chain

Let $\mathcal{G}_N$ be the set of isomorphism classes of connected cubic multigraphs on N vertices. A loop contributes two to the degree, and parallel edges are counted separately.

Let $e = uv$ be a non-loop edge. Besides e, two half-edges a, b are incident with u and two half-edges c, d are incident with v. The current local partition is

$$\{a, b\} \mid \{c, d\}.$$

Collapsing e and expanding again gives the two other partitions

$$\{a, c\} \mid \{b, d\}, \qquad \{a, d\} \mid \{b, c\}.$$

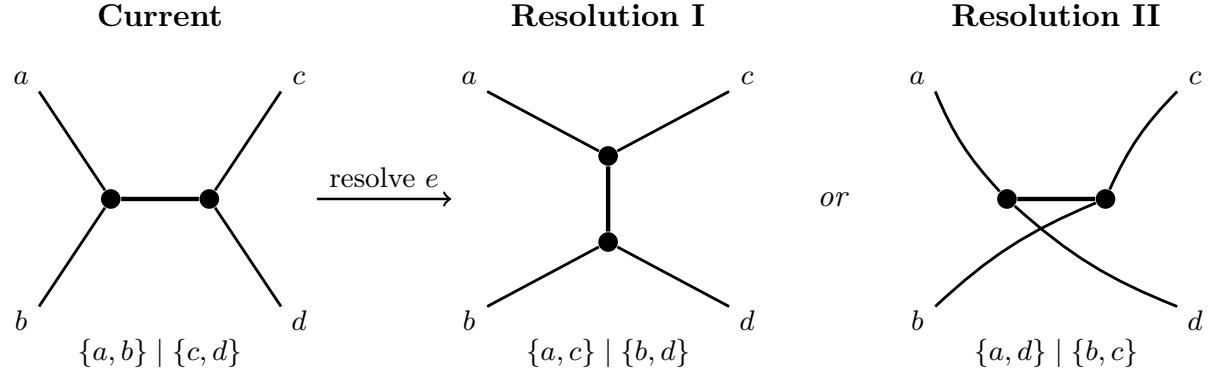


Figure 1: The local move at a non-loop edge. The central edge is retained, while the four other incident half-edges are repartitioned between its two endpoints. There are exactly three partitions of a four-element set into two unordered pairs: the current partition and the two resolutions.

These define the two Whitehead moves $w_e^{(1)}$ and $w_e^{(2)}$ on e; see Figure 1. The symbols a, b, c, d denote half-edges; two of them may belong to the same edge when loops or multiple edges are present.

If e is a loop, we define both $w_e^{(1)}$ and $w_e^{(2)}$ to be the identity, as in [3]. Thus every graph has $3N$ Whitehead moves, indexed by an edge and one of the two choices on it. These moves are counted separately even if they produce isomorphic graphs or leave the isomorphism class unchanged. For $G, H \in \mathcal{G}_N$, let m_{GH} be the number of moves taking a representative of G to a graph isomorphic to H. This number does not depend on the representative, and

$$\sum_{H \in \mathcal{G}_N} m_{GH} = 3N.$$

Definition 2.1 (Local Whitehead chain)**.** The transition matrix P_N on $\mathcal{G}_N$ is

$$P_N(G, H) = \frac{m_{GH}}{3N}. \tag{2.1}$$

Proposition 2.2 (Connected support)**.** *The support of P_N is connected for every even $N \geq 2$.*

Proof. This is the connectivity assertion in [3, Theorem 2.2], proved there with the same move and loop conventions. We include the reduction because connectivity is also what makes the enumeration in Section 8 exhaustive. For $N = 2$, the two classes are adjacent, as the two-state chain below shows.

Suppose $N \geq 4$. If the non-loop edges contain a cycle, choose a shortest one, of length $k \geq 2$. It has no chord. At one of its edges, one resolution places the two neighboring cycle half-edges at the same endpoint of the central edge. The cycle is thereby shortened from length k to length $k - 1$; when $k = 2$, the remaining edge becomes a loop. Continuing along the shortened cycle turns it into a loop after $k - 1$ moves. Each completed reduction creates one loop and removes none. Since a cubic graph has at most one loop at each vertex, repeated reductions eventually leave no non-loop cycle. The non-loop edges are then acyclic and connected, since loops do not join different vertices. They therefore form a tree, and every leaf of that tree carries a loop. There are $3N/2 - (N - 1) = N/2 + 1$ loops, so all resulting rooted trees have the same number of leaves.

It remains to connect all such looped trees. Choose a leaf v, delete all loops, delete v and its remaining incident edge, and mark the newly exposed vertex as the root. The result is a rooted full binary tree: after orienting its edges away from the root, every non-leaf has two children. Resolving

an internal edge performs an ordinary tree rotation. Give the two children at each vertex an arbitrary left–right order. If the left child of the root is not a leaf, rotate its edge with the root. The new left subtree is a proper subtree of the old one, so repetition eventually makes the left child a leaf. Now leave that leaf in place and apply the same procedure inside the right subtree. Induction on the number of internal vertices produces the right comb, in which every internal vertex has a leaf as its left child except at the final cherry. After the left–right order is forgotten, this comb depends only on its number of leaves. Restoring v and all the loops therefore gives a fixed looped cubic tree determined only by N. Hence every class can be joined to that class. □

Thus P_N is irreducible. Since $\mathcal{G}_N$ contains graphs with loops and the moves on a loop leave the state unchanged, it is also aperiodic. It therefore has a unique stationary distribution. In general $m_{GH} \neq m_{HG}$, so this distribution need not be uniform.

The two states at $N = 2$ already show why multiplicity matters. Write G_0 for the triple edge and G_1 for the graph consisting of one joining edge and one loop at each endpoint; see Figure 2. Directly counting the six moves from each graph gives

$$(m_{G_iG_j}) = \begin{pmatrix} 3 & 3 \\ 2 & 4 \end{pmatrix}, \qquad P_2 = \begin{pmatrix} 1/2 & 1/2 \\ 1/3 & 2/3 \end{pmatrix}.$$

The stationary probabilities are 2/5 and 3/5, not 1/2 and 1/2.

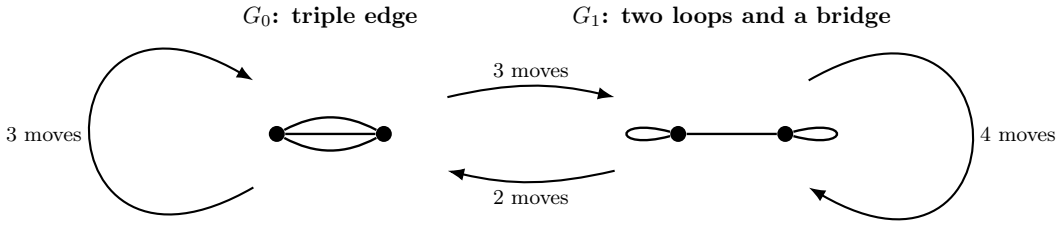


Figure 2: The chain for $N = 2$. Arrow labels give the numbers of Whitehead moves between the two classes, including moves that leave the class unchanged. The transition multiplicities in the two directions differ.

3 Configuration Space and Stationary Measure

To determine the stationary distribution, give each vertex $i \in [N]$ three distinguishable half-edges $(i, 1), (i, 2), (i, 3)$. A *configuration* is a perfect matching of these $3N$ half-edges. Contracting each triple to one vertex turns a configuration into a labelled cubic multigraph. A matched pair within one triple becomes a loop.

For a labelled cubic multigraph G on $[N]$, let m_{ii} be the number of loops at i and m_{ij} the number of edges joining i and j for $i < j$. Define

$$c(G) = \frac{(3!)^N}{\prod_i 2^{m_{ii}} m_{ii}! \prod_{i<j} m_{ij}!}. \tag{3.1}$$

Let $\mathrm{Aut}_v(G)$ be the group of vertex permutations preserving all adjacency multiplicities.

Lemma 3.1 (Counting configurations)**.** *A fixed labelled cubic multigraph G is represented by exactly $c(G)$ half-edge configurations. Consequently, the number of configurations whose unlabelled graph lies in the class $G \in \mathcal{G}_N$ is*

$$\frac{N!}{|\operatorname{Aut}_v(G)|}\, c(G). \tag{3.2}$$

Proof. At each vertex there are 3! ways to assign its three labelled half-edges to the three incidences of the adjacency multigraph, giving $(3!)^N$ assignments in all. Each matching is counted twice for every loop, $m_{ii}!$ times for the ordering of the loops at vertex i, and $m_{ij}!$ times for the ordering of the parallel edges between i and j. Dividing by these factors gives (3.1).

There are $N!/|\operatorname{Aut}_v(G)|$ labelled adjacency multigraphs in the unlabelled class of G. Multiplication by $c(G)$ proves (3.2). □

We next define the move on configurations. Regard each matched pair as an edge whose ends occupy labelled positions at their incident vertices. If the chosen edge joins distinct vertices u and v, choose one of the other two edge ends at u and one at v, and exchange their positions. There are four choices. In the notation of Figure 1, exchanging a with c or b with d gives one resolution, while exchanging a with d or b with c gives the other. If the chosen edge is a loop, all four exchanges are defined to be the identity. There are therefore $(3N/2) \cdot 4 = 6N$ exchanges from each configuration, counted with multiplicity.

Lemma 3.2 (Symmetric lift)**.** *The lifted configuration kernel is symmetric. Its restriction to connected configurations is closed, and after projection to connected isomorphism classes it is the kernel P_N of* (2.1)*.*

Proof. Each exchange is an involution: its reverse exchanges the same two edge ends. Hence the number of lifted exchanges from a configuration x to a configuration y equals the number from y to x. Uniform measure on configurations is therefore reversible. Relabelling vertices or permuting the three positions at a vertex commutes with the lifted transition, so its projection is a Markov kernel on isomorphism classes.

Connectedness is preserved: collapse the chosen edge to obtain a connected quotient, then expand the four-valent vertex into two vertices still joined by the central edge. The connected configurations therefore form a closed subset.

For a non-loop edge, each of the two Whitehead moves has two realizations among the four exchanges. Its projected probability is therefore $2/(6N) = 1/(3N)$. At a loop, all four exchanges leave the graph unchanged, giving the same total probability as the two trivial Whitehead moves. Summing over moves with isomorphic results gives $m_{GH}/(3N)$. □

Theorem 3.3 (Stationary law)**.** *For all $G, H \in \mathcal{G}_N$,*

$$\frac{c(G)}{|\operatorname{Aut}_v(G)|} m_{GH} = \frac{c(H)}{|\operatorname{Aut}_v(H)|} m_{HG}. \tag{3.3}$$

Consequently, the local Whitehead chain is reversible with stationary law

$$\pi_N(G) = \frac{c(G)/|\operatorname{Aut}_v(G)|}{\displaystyle\sum_{K \in \mathcal{G}_N} c(K)/|\operatorname{Aut}_v(K)|}. \tag{3.4}$$

This is the cubic configuration model conditioned on connectedness and projected to isomorphism classes.

Proof. By Lemma 3.2, uniform measure conditioned on connected configurations is reversible and projects to P_N. The mass of an unlabelled fibre is given by (3.2). The common factor $N!$ cancels on normalization, which gives (3.4). Summing detailed balance on configurations over the fibres of G and H gives (3.3). □

Remark 3.4. Automorphism order alone does not determine the stationary weight. In the $N = 2$ example, both graphs have vertex-automorphism group of order two, but $c(G_0) = 6$ and $c(G_1) = 9$, producing the stationary ratio $2 : 3$.

4 Pairing-Model Estimates

By Theorem 3.3, we may use the pairing model to estimate stationary probabilities on $\mathcal{G}_N$. This model was introduced by Bollobás [1]; see also the survey [12]. We shall use the following estimates.

Lemma 4.1 (Configuration estimates)**.** *Let $\widehat{G}_N$ be the cubic multigraph obtained from a uniform pairing of $3N$ half-edges in N labelled boxes of size three, without conditioning on connectedness.*

1. $\mathbb{P}(\widehat{G}_N$ *is disconnected*$) = O(N^{-1})$.
2. *Conditional on connectedness, the number L of loops converges in distribution and in every fixed moment to* Pois(1).
3. *Conditional on connectedness, the probability of a bridge whose smaller side has at least three vertices is $O(N^{-1})$.*

Proof. Disconnected configurations. Write $\mathcal{C}_N$ for the event that $\widehat{G}_N$ is connected. We first bound $\mathbb{P}(\mathcal{C}_N^c)$. If a set of $k \le N/2$ vertices is a union of connected components, then all $3k$ half-edges in the set are paired internally; in particular, k is even. The expected number of such k-sets is

$$D_{N,k} = \binom{N}{k} \frac{(3k-1)!!(3N-3k-1)!!}{(3N-1)!!}. \tag{4.1}$$

Here and below $(-1)!! = 1$. The term $k = 2$ can be read off directly:

$$D_{N,2} = \binom{N}{2} \frac{5!!}{(3N-1)(3N-3)(3N-5)} = \frac{5}{18N} + O(N^{-2}).$$

We give the uniform estimate used to sum the remaining terms. There are absolute constants $a, b > 0$ such that, for every $m \ge 1$,

$$a\sqrt{m}\,(m/e)^m \le m! \le b\sqrt{m}\,(m/e)^m. \tag{4.2}$$

For even $M \ge 2$, the identity $(M-1)!! = M!/(2^{M/2}(M/2)!)$ therefore gives

$$a_0(M/e)^{M/2} \le (M-1)!! \le b_0(M/e)^{M/2} \tag{4.3}$$

with absolute positive constants a_0, b_0. The same bounds give, uniformly for $1 \le k < N$,

$$\binom{N}{k} \le C_0 \sqrt{\frac{N}{k(N-k)}}\, \exp\{NH(k/N)\}, \qquad H(x) = -x\log x - (1-x)\log(1-x).$$

Applying these inequalities to (4.1) and collecting the powers of k, $N-k$, and N gives, for even $4 \le k \le N/2$,

$$D_{N,k} \le C\sqrt{\frac{N}{k(N-k)}}\exp\left\{-\frac{N}{2}H(k/N)\right\}. \tag{4.4}$$

Indeed, writing $x = k/N$, the powers from the double factorials satisfy

$$\frac{(3k)^{3k/2}(3N-3k)^{(3N-3k)/2}}{(3N)^{3N/2}} = x^{3Nx/2}(1-x)^{3N(1-x)/2} = \exp\{-3NH(x)/2\}.$$

The binomial coefficient contributes $\exp\{NH(x)\}$, leaving the exponential and square-root prefactor in (4.4).

When $k \le N/4$, the inequality $H(k/N) \ge (k/N)\log(N/k)$ and $N-k \ge N/2$ reduce the right-hand side to

$$u_k = Ck^{-1/2}(k/N)^{k/2}.$$

For even $k \ge 4$ with $k+2 \le N/4$,

$$\frac{u_{k+2}}{u_k} = \sqrt{\frac{k}{k+2}}\frac{k+2}{N}\left(1+\frac{2}{k}\right)^{k/2} \le \frac{e}{4} < 1.$$

Thus the small-k terms form a geometric tail beginning with $u_4 = O(N^{-2})$. In the range $N/4 \le k \le N/2$, the entropy is at least $H(1/4)$; even after summing $O(N)$ terms, the contribution is exponentially small. A union bound now proves $\mathbb{P}(\mathcal{C}_N^c) = O(N^{-1})$.

Loops. A vertex supports at most one loop in a cubic configuration. To specify an ordered r-tuple of loops, choose r distinct vertices, choose one of the three pairs of half-edges at each vertex, and require those r pairs to occur in the random matching. Therefore, for fixed r,

$$\mathbb{E}(L)_r = (N)_r 3^r \prod_{j=0}^{r-1}\frac{1}{3N-(2j+1)} \longrightarrow 1. \tag{4.5}$$

These are the limiting factorial moments of Pois(1), a moment-determinate law. Since every power L^q is a fixed linear combination of $(L)_0, \dots, (L)_q$, (4.5) also gives $\sup_N \mathbb{E}L^q < \infty$ for every fixed q. To justify conditioning, fix q and use Cauchy–Schwarz:

$$\mathbb{E}\big[L^q \mathbf{1}_{\mathcal{C}_N^c}\big] \le \big(\mathbb{E}L^{2q}\big)^{1/2}\mathbb{P}(\mathcal{C}_N^c)^{1/2} = o(1).$$

It follows that

$$|\mathbb{E}[L^q \mid \mathcal{C}_N] - \mathbb{E}L^q| = o(1).$$

Moreover, for every event E determined by L,

$$|\mathbb{P}(E \mid \mathcal{C}_N) - \mathbb{P}(E)| \le \frac{2\mathbb{P}(\mathcal{C}_N^c)}{\mathbb{P}(\mathcal{C}_N)} = o(1).$$

Conditioning therefore preserves both the distributional limit and every fixed moment.

Large bridge sides. Suppose a bridge cuts off a vertex set S of size $k \le N/2$. The degree sum on S gives $3k = 2|E(S)| + 1$, so k is odd. To count candidate sets, choose one crossing half-edge on each side and pair all remaining half-edges internally. For odd $k \ge 3$, the expected number of k-sets with exactly one crossing pair is therefore

$$A_{N,k} = \binom{N}{k}(3k)(3N-3k)\frac{(3k-2)!!(3N-3k-2)!!}{(3N-1)!!}. \tag{4.6}$$

The first term is

$$A_{N,3} = \frac{35}{18N} + O(N^{-2}).$$

Apply (4.2)–(4.3) to (4.6). Replacing $3k-1$ by $3k$, and $3(N-k)-1$ by $3(N-k)$, changes the result by bounded factors. After cancellation, the upper bound is

$$C\sqrt{\frac{N}{k(N-k)}}e^{NH(k/N)}k(N-k)\frac{(3k)^{(3k-1)/2}(3N-3k)^{(3N-3k-1)/2}}{(3N)^{3N/2}}.$$

The powers of k and $N-k$ outside the exponential reduce its prefactor to $C\sqrt{N}$, while the entropy exponent is again $NH - 3NH/2$. Thus

$$A_{N,k} \le C\sqrt{N}\exp\left\{-\frac{N}{2}H(k/N)\right\}, \qquad 3 \le k \le N/2. \tag{4.7}$$

For $k \le N/4$, this is at most $v_k = C\sqrt{N}(k/N)^{k/2}$. If $k+2 \le N/4$, then

$$\frac{v_{k+2}}{v_k} = \frac{k+2}{N}\left(1+\frac{2}{k}\right)^{k/2} \le \frac{e}{4}.$$

The odd tail beginning at $k=5$ is therefore $O(v_5) = O(N^{-2})$. In the central range, summing (4.7) gives $O(N^{3/2}e^{-NH(1/4)/2})$. Together with the displayed value of $A_{N,3}$, this proves that the sum is $O(N^{-1})$. The sum of the $A_{N,k}$ is an upper bound for the probability in question, because the internally paired sides counted in (4.6) need not themselves be connected. Conditioning on $\mathcal{C}_N$, whose probability is $1 - O(N^{-1})$, changes the bound only by a factor $1 + O(N^{-1})$. □

By Theorem 3.3, these conditional laws are precisely the laws of the corresponding statistics under π_N.

5 The Local Spectrum

Since the chain is reversible, P_N is self-adjoint on $L^2(\pi_N)$. We obtain a lower bound for its spectrum by considering the four exchanges at a fixed edge.

Proposition 5.1 (Spectral interval)**.** *For every even N,*

$$\mathrm{Spec}(P_N) \subseteq [-1/2, 1].$$

Proof. Let $\widetilde{\Omega}_N$ be the following labelled cover of the connected configuration space. Fix $3N/2$ labelled abstract edges, give each edge two distinguishable ends, and assign the $3N$ ends bijectively to three labelled positions at each of N labelled vertices, retaining the assignments whose induced multigraph is connected. Equip this finite set with uniform measure. Every connected configuration has exactly

$$2^{3N/2}(3N/2)!$$

preimages, obtained by labelling and orienting its edges. If q sends a point of $\widetilde{\Omega}_N$ to its unlabelled graph class, then the pushforward of uniform measure by q is π_N. Consequently

$$U : L^2(\pi_N) \longrightarrow L^2(\widetilde{\Omega}_N), \qquad (UF)(\omega) = F(q(\omega)),$$

is an isometric embedding.

Fix a labelled edge e. Define K_e on the cover as follows. If the two ends of e occupy different vertex boxes, choose one of the two remaining positions in each box and exchange the edge ends at those positions; K_e averages the four possible exchanges. Each exchange is an involution, so K_e is self-adjoint for uniform measure. If e is a loop, set K_e to be the identity.

For a function in the range of U, condition on the positions of the two ends of e and on all positions outside its endpoint boxes. On the event that e is not a loop, these data divide the cover into fibres. Within one fibre, the four remaining edge ends can be assigned to the four remaining positions in 4! ways. Regard these positions as the vertices of $K_{2,2}$, with one part in each endpoint box. The allowed exchanges are the transpositions along its four edges. Transpositions along the edges of any connected graph generate the full symmetric group on its vertices, so these four exchanges connect all 4! assignments. Starting from a connected assignment, every assignment reached in this way remains connected because each exchange is a Whitehead move.

Label the four remaining ends $1, 2, 3, 4$. A function on isomorphism classes is invariant under permutations of the two positions within either box, so its value depends only on the two-subset placed in the first box. The six possibilities are

$$12,\ 13,\ 14,\ 23,\ 24,\ 34.$$

Each has four preimages among the 4! assignments and therefore has uniform measure. An exchange replaces one member of the two-subset by one member of its complement. Thus two states are adjacent exactly when their intersection has size one. The induced operator is simple random walk on the Johnson graph $J(4, 2)$, which is the graph of an octahedron.

Let A be its adjacency matrix. The three complementary pairs are $12 \leftrightarrow 34$, $13 \leftrightarrow 24$, and $14 \leftrightarrow 23$. Let C be the involution that exchanges each pair. Then

$$A = J - I - C,$$

where J is the all-ones operator. The constant vector has eigenvalue 4. On its orthogonal complement $J = 0$, so the $+1$ and -1 eigenspaces of C give eigenvalues -2 and 0, respectively. The $+1$ eigenspace has dimension three and contains the constant vector; the -1 eigenspace has dimension three. Hence $A/4$ has eigenvalues

$$1 \quad \text{(once)}, \qquad -\tfrac{1}{2} \quad \text{(twice)}, \qquad 0 \quad \text{(three times)}.$$

The loop fibres contribute only the eigenvalue 1. Averaging the conditional quadratic forms over all fibres gives, for every $F \in L^2(\pi_N)$,

$$-\tfrac{1}{2}\langle UF, UF\rangle \le \langle UF, K_e UF\rangle \le \langle UF, UF\rangle.$$

The lifted transition operator is

$$\widetilde{P}_N = \frac{2}{3N} \sum_e K_e.$$

For a non-loop e, two of its four exchanges give each Whitehead resolution; for a loop, all four are the identity. Thus the definition of P_N gives the intertwining identity $\widetilde{P}_N U = U P_N$. Averaging the preceding bounds over e and using that U is an isometry yields

$$-\tfrac{1}{2}\|F\|^2_{\pi_N} \le \langle F, P_N F\rangle_{\pi_N} \le \|F\|^2_{\pi_N}.$$

Since P_N is self-adjoint, the Rayleigh-quotient characterization of its extreme eigenvalues proves the proposition. □

We next use the number of loops to bound eigenvalues near 1.

6 Loop Count and Low Eigenvalues

For a reversible kernel, write

$$\mathcal{E}_N(F,F) = \langle F, (I-P_N)F\rangle_{\pi_N} = \frac{1}{2}\sum_{G,H} \pi_N(G)P_N(G,H)\big(F(G)-F(H)\big)^2. \tag{6.1}$$

Let $L(G)$ denote the number of loops in G, and put $\mu_N(k) = \pi_N\{G : L(G) = k\}$.

Lemma 6.1 (Moves that remove a loop)**.** *Let $N \geq 4$, and let G have k loops. Exactly $2k$ Whitehead moves lead to a graph with $k-1$ loops. Every move changes the number of loops by at most one.*

Proof. A vertex u carrying a loop ℓ has one remaining incident edge s, its *stem*. Write v for the other endpoint of s, and c, d for the two noncentral ends at v. The two resolutions of s separate the two ends of ℓ and therefore destroy it; see Figure 3. They destroy a second loop only if c and d are paired to each other. In that case both u and v carry loops and s uses their remaining degrees, so connectedness forces $N = 2$. Nor can either resolution create a loop: such a loop would have to come from an edge parallel to s, but the two noncentral ends at u are already paired to form ℓ.

No other chosen edge can destroy ℓ: a move on ℓ is trivial, and only its stem can move one of its ends to another vertex. Distinct loop vertices have distinct stems when $N \geq 4$, since a shared stem would again be the entire two-vertex component. Thus the two stem resolutions associated with each loop are distinct and give exactly $2k$ moves that remove a loop.

Finally, a resolution can create a loop only by putting the two ends of an edge parallel to the chosen central edge into the same new vertex. Creating two loops at once would require both pairs of noncentral ends to be paired across the two central vertices. Those two edges together with the central edge would form an isolated triple edge, again forcing $N = 2$. Applying the same argument to the inverse resolution also rules out the destruction of two loops. Hence $|L(G') - L(G)| \leq 1$. □

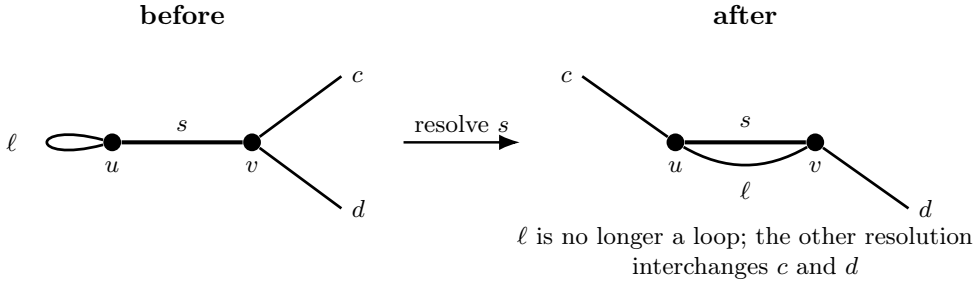


Figure 3: Resolving the stem s of a loop ℓ. In either resolution the two ends of ℓ move to different vertices, while the two other incident half-edges separate. Thus each loop contributes exactly two moves that decrease the number of loops.

Theorem 6.2 (Loop-count Dirichlet form)**.** *For $N \geq 4$ and every function f on the nonnegative integers,*

$$\mathcal{E}_N(f(L), f(L)) = \frac{2}{3N}\sum_{k\geq 1} k\mu_N(k)\big(f(k)-f(k-1)\big)^2. \tag{6.2}$$

Consequently, if $0 = \lambda_1 \leq \lambda_2 \leq \cdots$ are the eigenvalues of $I - P_N$, then for every fixed $r \geq 1$,

$$\lambda_{r+1} \leq \frac{2r}{3N} + o(N^{-1}). \tag{6.3}$$

Proof. By Lemma 6.1, the stationary directed flow from level k to level $k-1$ is

$$\mu_N(k)\frac{2k}{3N}.$$

Detailed balance says that the reverse aggregate flow has the same value. The two directions of each transition cancel the factor $1/2$ in (6.1). Summing over k proves (6.2).

It remains to choose a test space. Let ν be the Pois(1) law and define the Charlier polynomials C_j by the generating function [5]

$$\sum_{j\ge 0} C_j(k)\frac{t^j}{j!} = e^{-t}(1+t)^k. \tag{6.4}$$

If $X \sim \nu$, orthogonality follows directly by multiplying two generating functions:

$$\mathbb{E}\big[e^{-s}(1+s)^X e^{-t}(1+t)^X\big] = e^{-s-t}\exp\big((1+s)(1+t)-1\big) = e^{st}.$$

Comparing coefficients, and subtracting the generating functions at k and $k-1$, gives

$$\mathbb{E}[C_i(X)C_j(X)] = j!\,\mathbf{1}_{\{i=j\}}, \qquad C_j(k) - C_j(k-1) = jC_{j-1}(k-1).$$

Since $k\nu(k) = \nu(k-1)$, the limiting form

$$\mathcal{D}(f,f) = \sum_{k\ge 1} k\nu(k)\big(f(k)-f(k-1)\big)^2$$

satisfies

$$\mathcal{D}(C_i, C_j) = j\,j!\,\mathbf{1}_{\{i=j\}}.$$

Fix r. For $0 \le i, j \le r$, define the Gram and difference matrices

$$B_N(i,j) = \sum_{k\ge 0} \mu_N(k)C_i(k)C_j(k),$$

and

$$D_N(i,j) = \sum_{k\ge 1} k\mu_N(k)\big(C_i(k)-C_i(k-1)\big)\big(C_j(k)-C_j(k-1)\big).$$

Every entry is the expectation of a fixed polynomial in L. Moment convergence in Lemma 4.1 therefore gives

$$B_N \longrightarrow B = \operatorname{diag}(0!, 1!, \dots, r!), \qquad D_N \longrightarrow D = \operatorname{diag}(0\,0!, 1\,1!, \dots, r\,r!).$$

The limiting Gram matrix B is positive definite. Hence B_N is positive definite for all sufficiently large N, the functions $C_0(L), \dots, C_r(L)$ are then linearly independent, and

$$\lambda_{\max}\big(B_N^{-1/2} D_N B_N^{-1/2}\big) \longrightarrow \lambda_{\max}\big(B^{-1/2} D B^{-1/2}\big) = r.$$

Equivalently, the largest quotient $x^{\mathsf{T}} D_N x/(x^{\mathsf{T}} B_N x)$ on their span is $r + o(1)$. By (6.2), the largest Laplacian Rayleigh quotient on that $(r+1)$-dimensional span is therefore $2r/(3N) + o(N^{-1})$. The Courant–Fischer min–max principle gives (6.3). □

For $r = 1$, one may take the centered test function $L - \mathbb{E}_{\pi_N} L$; its Rayleigh quotient is $2/(3N) + o(N^{-1})$. The higher Charlier polynomials provide the remaining test functions. The process $L(G_t)$ need not itself be Markov: transition probabilities between loop counts can depend on the rest of the graph. The identity (6.2) uses only the total stationary flow between successive values of L.

7 Bridges and Conductance

Let $\mathcal{B}_N \subset \mathcal{G}_N$ be the classes with at least one bridge, let $\mathcal{B}_0 = \mathcal{G}_N \setminus \mathcal{B}_N$, and let $\mathcal{B}_1$ be the classes with exactly one bridge. For sets $S, T \subseteq \mathcal{G}_N$, write

$$Q_N(S,T) = \sum_{G \in S,\, H \in T} \pi_N(G) P_N(G,H).$$

The conductance of the cut between bridged and bridgeless graphs is

$$\Phi_N(\mathcal{B}_N) = \frac{Q_N(\mathcal{B}_N, \mathcal{B}_0)}{\min\{\pi_N(\mathcal{B}_N), \pi_N(\mathcal{B}_0)\}}.$$

The bridge-crossing description underlies the bottleneck argument of [3, Section 3]. We give a proof using the criterion that an edge is a bridge if and only if it lies on no cycle.

Lemma 7.1 (Whitehead moves and bridges)**.** *Let G' be obtained from G by resolving an edge e.*

1. *Every bridge of G other than e remains a bridge of G'.*
2. *If e is the unique bridge of G, then either resolution of e is bridgeless.*

Consequently, a resolution crosses from $\mathcal{B}_N$ to $\mathcal{B}_0$ if and only if it is one of the two resolutions of the unique bridge of a graph in $\mathcal{B}_1$.

Proof. Let $f \neq e$ be a bridge of G, and consider the two components of $G - f$. The edge e lies entirely in one component. Collapsing and re-expanding e changes incidences only on that side of the cut: it creates no second edge between the two components. Thus f is still their only connection in G' and remains a bridge. This proves the first assertion and also shows that a crossing move must act on the unique bridge.

For the converse, suppose that $e = uv$ is the unique bridge, and let A and B be the components of $G - e$ containing u and v. Neither side contains a bridge: an edge separating A or B would also separate G. In one of the two resolutions, let $a = ux$ be the edge whose end at u moves to v, and let $b = vy$ be the edge whose end at v moves to u. Thus, after the resolution, the incidences of the three crossing edges are

$$e = uv, \qquad a = vx, \qquad b = uy.$$

The other resolution has the same description after choosing the other incident edge on each side. This notation also covers a loop: if a was a loop at u, then $x = u$, and moving one of its ends turns it into the edge vu.

Because a is not a bridge of A, the graph $A - a$ contains a path P_A from u to x; similarly, $B - b$ contains a path P_B from v to y. The three crossing edges lie on the following cycles in G':

$$e + a + P_A, \qquad e + b + P_B, \qquad a + P_B + b + P_A.$$

Here an expression records the edges and paths in their cyclic order; it also covers the case in which one of P_A, P_B has length zero.

It remains to check edges internal to the two sides. Let f be an edge of $A - a$. Since A is bridgeless, f lies on a cycle C in A. If C avoids a, it survives unchanged. If C uses the old edge $a = ux$, replace that edge in C by the path $u - e - v - a - x$. This gives a cycle through f in G'. The same argument in B, replacing the old $b = vy$ by $v - e - u - b - y$, treats every edge of $B - b$. Hence every edge of G' lies on a cycle, and G' is bridgeless. □

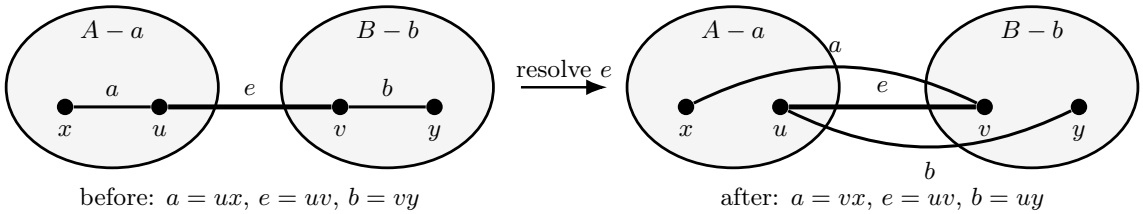


Figure 4: One resolution of the unique bridge $e = uv$. The edge $a = ux$ becomes $a = vx$, and $b = vy$ becomes $b = uy$. Since $A - a$ and $B - b$ remain connected, their internal paths close cycles through all three crossing edges and replace any side cycle that formerly used a or b.

Theorem 7.2 (Bridge flow)**.** *For every even $N \geq 2$,*

$$Q_N(\mathcal{B}_N, \mathcal{B}_0) = \frac{2}{3N}\pi_N(\mathcal{B}_1). \tag{7.1}$$

Moreover,

$$\pi_N(\mathcal{B}_0) \longrightarrow e^{-1}, \qquad \pi_N(\mathcal{B}_1) \longrightarrow e^{-1}, \qquad \pi_N(\mathcal{B}_N) \longrightarrow 1 - e^{-1}, \tag{7.2}$$

and hence

$$\Phi_N(\mathcal{B}_N) = \frac{2}{3N}(1 + o(1)). \tag{7.3}$$

Proof. By Lemma 7.1, exactly two moves take a graph in $\mathcal{B}_1$ to $\mathcal{B}_0$, whereas no move takes a graph with more than one bridge to $\mathcal{B}_0$. Each move has probability $1/(3N)$. Summing with respect to π_N gives (7.1).

In a connected cubic multigraph with $N \geq 4$, every loop vertex is joined to the rest of the graph by a bridge. Conversely, a bridge that cuts off a single vertex must have a loop at that vertex. Each side of a bridge has odd order, so, if no bridge has at least three vertices on its smaller side, these singleton-side bridges are all the bridges. By Lemma 4.1, the exceptional event has probability $O(N^{-1})$ under π_N. It follows that $\mathcal{B}_0$ differs from $\{L = 0\}$ by a set of stationary mass $O(N^{-1})$, and that $\mathcal{B}_1$ differs from $\{L = 1\}$ by such a set. The Poisson limit for L proves the first two limits in (7.2); the third follows because $\mathcal{B}_N$ is the complement of $\mathcal{B}_0$. Since $e^{-1} < 1 - e^{-1}$, $\mathcal{B}_0$ is the smaller side for all sufficiently large N. Substitution in (7.1) gives (7.3). □

Theorem 7.3 (Exact bridge conductance)**.** *For every even $N \geq 2$, writing $E_N = 3N/2$, we have*

$$(E_N - 1)\pi_N(\mathcal{B}_1) = E_N \pi_N(\mathcal{B}_0). \tag{7.4}$$

The bridgeless set $\mathcal{B}_0$ is the smaller side of the cut, and

$$\Phi_N(\mathcal{B}_N) = \frac{1}{E_N - 1}. \tag{7.5}$$

Proof. Let a_G be the order of the full incidence automorphism group, including loop flips and permutations of parallel edges. Thus

$$a_G = |\operatorname{Aut}_v(G)| \prod_i 2^{m_{ii}} m_{ii}! \prod_{i<j} m_{ij}!,$$

and the stationary weight is proportional to a_G^{-1}. Define $B(t) = \sum_G t^{|V(G)|}/a_G$ over connected bridgeless cubic graphs, and $D(t)$ over connected cubic graphs with exactly one bridge. Let $A(t)$

count connected cubic graphs with one external half-edge and no internal bridge, with automorphisms fixing that half-edge. Deleting the unique bridge gives an unordered pair of graphs counted by A. Removing the external half-edge and suppressing its vertex gives a bridgeless cubic graph with a distinguished edge, except for the one-loop graph of weight $t/2$. Accounting for symmetry factors gives

$$D = \frac{A^2}{2}, \qquad A = \frac{t}{2} + \frac{3}{2}t^2 B'. \tag{7.6}$$

All series below are formal. For a standard Gaussian variable X, set

$$W(t, J) = \log \mathbb{E} e^{tX^3/6 + JX}, \qquad \phi = W_J(t, J), \qquad \Gamma(t, \phi) = J\phi - W(t, J).$$

The combinatorial Legendre transform expresses Γ in terms of connected graphs without internal bridges; see [2, Section 5.8]. With the present sign convention its constant and linear terms are

$$\Gamma(t, 0) = -B(t), \qquad \Gamma_\phi(t, 0) = -A(t).$$

These terms include vacuum graphs and the one-loop one-point graph. For completeness, the cubic change of variables gives

$$\Gamma(t, \phi) = \frac{\phi^2}{2} - \frac{t\phi^3}{6} + \frac{1}{2}\log(1 - t\phi) - B\big(t(1 - t\phi)^{-3/2}\big). \tag{7.7}$$

Indeed, put $m = 1 - t\phi$, $s = tm^{-3/2}$, and $K = (J - \phi + t\phi^2/2)/\sqrt{m}$. Formal Gaussian translation and scaling give

$$e^{W(t,J)} = m^{-1/2} e^{-\phi^2/2 + t\phi^3/6 + J\phi} e^{W(s,K)}.$$

The condition $W_J(t, J) = \phi$ is equivalent to $W_J(s, K) = 0$, so $W(s, K) = B(s)$, proving (7.7). This identity is also given in [2, Section 7.5].

Twice differentiating (7.7) at $\phi = 0$ and using (7.6) yields

$$\Gamma_{\phi\phi}(t, 0) = 1 + \frac{t}{2}A - \frac{3t^2}{2}A'.$$

At the zero-mean source $J = -A(t)$, Gaussian integration by parts and the inverse-Hessian identity give

$$A = \frac{t}{2}\langle X^2 \rangle_{t, -A(t)} = \frac{t}{2\Gamma_{\phi\phi}(t, 0)}.$$

Here the brackets denote the normalized formal expectation. Consequently,

$$3tAA' - A^2 = \frac{2A}{t} - 1, \qquad \left(\frac{3}{2}t\frac{d}{dt} - 1\right) D = \frac{3}{2}tB'.$$

Taking the coefficient of t^N proves $(E_N - 1)[t^N]D = E_N[t^N]B$, hence (7.4). Bridgeless cubic graphs exist at every even order, for instance a cycle with a perfect matching added. Since $\mathcal{B}_1 \subseteq \mathcal{B}_N$ and $E_N/(E_N - 1) > 1$, the bridgeless side is strictly smaller. The exact flow formula (7.1) now gives (7.5). □

We use the usual worst-case total-variation mixing time

$$t_{\mathrm{mix}}(\varepsilon) = \min\big\{t : \max_G \|P_N^t(G, \cdot) - \pi_N\|_{\mathrm{TV}} \le \varepsilon\big\}.$$

We define the relaxation time by $t_{\mathrm{rel}} = 1/\lambda_2(I - P_N)$.

Corollary 7.4 (Mixing lower bound)**.** *For every even $N \geq 2$ and $0 < \varepsilon < 1$,*

$$t_{\mathrm{mix}}(\varepsilon) \geq (3N/2 - 1)\big(1 - \pi_N(\mathcal{B}_0) - \varepsilon\big). \tag{7.8}$$

In particular, for every fixed $\varepsilon < 1 - e^{-1}$,

$$t_{\mathrm{mix}}(\varepsilon) \geq \frac{3N}{2}\big(1 - e^{-1} - \varepsilon\big)(1 + o(1)). \tag{7.9}$$

In particular,

$$t_{\mathrm{mix}}(1/4) \geq 0.57318\ldots\ N(1 + o(1)).$$

The relaxation time is also at least $(3/2 + o(1))N$.

Proof. Start the chain from $\mu_0 = \pi_N(\cdot \mid \mathcal{B}_0)$. Pointwise, $\mu_0 \leq \pi_N/\pi_N(\mathcal{B}_0)$, and applying a Markov kernel preserves this domination. If $\mu_s = \mu_0 P_N^s$, the probability of crossing from $\mathcal{B}_0$ to $\mathcal{B}_N$ at step $s + 1$ is therefore

$$\sum_{G \in \mathcal{B}_0} \mu_s(G) P_N(G, \mathcal{B}_N) \leq \frac{Q_N(\mathcal{B}_0, \mathcal{B}_N)}{\pi_N(\mathcal{B}_0)}.$$

A union bound over the first t steps gives

$$\mu_0 P_N^t(\mathcal{B}_0) \geq 1 - t\frac{Q_N(\mathcal{B}_0, \mathcal{B}_N)}{\pi_N(\mathcal{B}_0)} = 1 - t\Phi_N(\mathcal{B}_N)$$

for every even $N \geq 2$; the equality uses reversibility and Theorem 7.3. Testing total variation on $\mathcal{B}_0$ gives

$$\|\mu_0 P_N^t - \pi_N\|_{\mathrm{TV}} \geq 1 - t\Phi_N(\mathcal{B}_N) - \pi_N(\mathcal{B}_0).$$

Using (7.5) proves (7.8), and the limit in (7.2) gives (7.9). Finally, convexity of total variation gives

$$\|\mu_0 P_N^t - \pi_N\|_{\mathrm{TV}} \leq \sum_G \mu_0(G)\|P_N^t(G, \cdot) - \pi_N\|_{\mathrm{TV}} \leq \max_G \|P_N^t(G, \cdot) - \pi_N\|_{\mathrm{TV}},$$

so the lower bound obtained from the conditional initial law also holds for worst-case mixing. The relaxation claim follows from (6.3) with $r = 1$: $\lambda_2(I - P_N) \leq 2/(3N) + o(N^{-1})$, and the reciprocal is the relaxation time. See [6] for the standard mixing and conductance definitions. □

8 Enumeration

For fixed N, we enumerate the isomorphism classes in $\mathcal{G}_N$ and compute the integer matrix $M_N = (m_{GH})$ by a breadth-first search. Every Whitehead move is counted, including moves with isomorphic results and those that leave the class unchanged. The tables below are computed from these matrices.

8.1 Canonical Representation

Store a cubic multigraph by its upper-triangular adjacency-multiplicity matrix, with diagonal entries counting loops. To compare two such matrices, form a colored simple incidence graph. Its first color class consists of the N original vertices. Every non-loop edge becomes a vertex of a second color, adjacent to its two endpoints. Every loop becomes a vertex of a third color, adjacent to its single incident original vertex. Keeping loops and non-loop edges in different colors retains the information lost by replacing a loop's double incidence with one simple incidence. Parallel edges remain distinct incidence vertices.

Two multigraphs are isomorphic if and only if their colored incidence graphs are color-preservingly isomorphic. Canonical representation is a standard way to avoid duplicate isomorphism classes in exhaustive generation [8]. We use nauty/Traces [9] to canonically label the incidence graph. Reading the original-vertex color class in canonical order and writing the corresponding upper triangle produces a canonical code. Two codes are equal exactly when the corresponding multigraphs are isomorphic.

8.2 Whitehead Moves and Breadth-First Search

The program represents each of the $3N/2$ edges by a pair of half-edges and records the vertex incident with each half-edge. For a non-loop edge $e = uv$, let s_u, s_v be its half-edges, let a, b be the other half-edges at u, and let c, d be the other half-edges at v. The pairing is unchanged by a Whitehead move. The two resulting graphs are obtained by changing the vertex incidences as follows:

a, c are incident with u,	b, d are incident with v;
a, d are incident with u,	b, c are incident with v.

The half-edges s_u, s_v remain at u, v. Converting the resulting incidence map back to an adjacency-multiplicity matrix implements exactly the two partitions in Figure 1, including cases in which two of a, b, c, d belong to the same edge.

Algorithm 8.1 (Transition enumeration)**.** Fix an even N and a connected cubic starting graph. Maintain a set $\mathcal{S}$ of canonical codes and a first-in, first-out queue, initially containing only the starting code. While the queue is nonempty:

1. remove one code and reconstruct its adjacency multigraph;
2. expand its $3N/2$ edges into paired half-edges, keeping parallel edges distinct;
3. for a loop, increase the diagonal entry of M_N by two; for a non-loop edge, perform the two Whitehead moves described above;
4. canonicalize each non-loop target, increment its transition multiplicity, and enqueue its code if it is not already in $\mathcal{S}$.

Return $\mathcal{S}$ and the integer multiplicity matrix $M_N = (m_{GH})$.

The search starts from the triple edge when $N = 2$. For $N \geq 4$, take a path $v_1 v_2 \cdots v_N$, put a loop at each endpoint, and, for $1 \leq i < N$, give $v_i v_{i+1}$ multiplicity one when i is odd and two when i is even. Adjacent path edges then have total multiplicity three at every internal vertex, while each endpoint has its loop and one path edge.

Proposition 8.2 (Correctness of the enumeration)**.** *Algorithm 8.1 terminates and returns every class in $\mathcal{G}_N$ exactly once. For every ordered pair of returned classes, its matrix entry is the number of Whitehead moves between them.*

Proof. There are finitely many adjacency-multiplicity matrices with N vertices and degree three, so the queue terminates. Canonical codes identify precisely the isomorphism classes, which proves that no class is stored twice. Every generated graph is cubic because u and v each remain incident with three half-edges. It is connected because collapsing e gives a connected graph and either displayed expansion reconnects the two new vertices by e. The Whitehead support on $\mathcal{G}_N$ is connected by

Proposition 2.2; hence a shortest support path from the starting class to any class is discovered inductively by the breadth-first search. Finally, the inner loop visits each edge and both Whitehead moves on it. Moves with the same resulting class are counted separately, giving the multiplicities in (2.1). □

Let $C_N = |\mathcal{G}_N|$, and let $T_{\mathrm{can}}(N)$ denote the maximum time for one canonical labelling of the colored incidence graphs used here. Such a graph has $5N/2$ vertices and at most $3N$ incidence edges. The search generates exactly $3NC_N$ Whitehead moves and makes at most $3NC_N + 1$ canonical-labelling calls, including the starting code. For each of the C_N states, constructing and copying the dense $N \times N$ target matrices for $3N$ moves costs $O(N^3)$ time. Canonical labelling contributes at most $3NT_{\mathrm{can}}(N)$ per state. The total time is

$$O\big(C_N\big(N^3 + NT_{\mathrm{can}}(N)\big)\big)\,.$$

Storing a dense code for each state costs $O(N^2C_N)$ machine words. If transitions are stored as pairs of state indices with multiplicities, they use $O(NC_N)$ more, giving $O(N^2C_N)$ total storage. The supplied program instead retains a separate record for every Whitehead move, including its target code and vertex permutations. Its storage bound is therefore $O(N^3C_N)$.

8.3 Computational Results

All counts are computed over the integers. For each N, the number of classes agrees with OEIS A005967 [10], every row of M_N sums to $3N$, and every transition between distinct classes has a reverse. Stationary weights are obtained by propagating the rational identities $w_G m_{GH} = w_H m_{HG}$ along the connected support and checking them on every edge. Normalizing these weights gives the stationary masses, cut flow, and conductance as reduced fractions. This calculation uses only M_N, not the formula in Theorem 3.3; it checks the table's stationary quantities without separately verifying the automorphism factors in that formula.

The bridges of each graph are found by deleting each edge and testing connectedness. These counts confirm Lemma 7.1: a class with one bridge has exactly two moves into $\mathcal{B}_0$, and a class with more than one bridge has none. The conductance is computed from the transition matrix before comparison with (7.5).

The computations used the native Traces backend with one worker on an Apple M5 computer with 24 GB of memory. At $N = 14$, the search visited 21,096 classes and all 886,032 Whitehead moves. The accompanying record contains the counts and rational values, running times, software versions, source hashes, and SHA-256 digests of the ordered canonical codes and sparse transition matrix. The full matrices can be reconstructed by the supplied programs. The tables are generated from this record. In Table 1, support edges are unordered pairs of distinct classes joined by a Whitehead move.

The last column of Table 2 verifies Theorem 7.3 for all seven enumerated sizes. The corresponding configuration-count identity is obtained by setting

$$W_i = \sum_{G \in \mathcal{B}_i} \frac{N!}{|\operatorname{Aut}_v(G)|} c(G), \qquad i = 0, 1.$$

By Lemma 3.1, W_i is the number of connected half-edge configurations whose unlabelled graph has i bridges. Then

$$(3N/2 - 1)W_1 = (3N/2)W_0. \tag{8.1}$$

This is (7.4) multiplied by the common stationary normalizing factor, and holds for every even $N \geq 2$.

Table 1: Enumeration sizes and running times. Times include generation and verification with one worker on the machine described in the text; table rendering is excluded. The first row also includes loading the native library.

N	$\lvert\mathcal{G}_N\rvert$	support edges	moves	seconds
2	2	1	12	< 0.01
4	5	4	60	< 0.01
6	17	25	306	< 0.01
8	71	179	1,704	0.09
10	388	1,537	11,640	0.56
12	2,592	15,402	93,312	5.90
14	21,096	177,630	886,032	68.70

Table 2: Stationary bridge data computed from the full transition matrices. Here $\mathcal{B}_0$ is the bridgeless set, $\mathcal{B}_1$ is the one-bridge set, and $\mathcal{B}_N$ is the set of all bridged classes.

N	$\pi_N(\mathcal{B}_0)$	$\pi_N(\mathcal{B}_1)$	$Q_N(\mathcal{B}_N, \mathcal{B}_0)$	$\Phi_N(\mathcal{B}_N)$
2	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{1}{5}$	$\frac{1}{2}$
4	$\frac{1}{3}$	$\frac{2}{5}$	$\frac{1}{15}$	$\frac{1}{5}$
6	$\frac{352}{1\,105}$	$\frac{396}{1\,105}$	$\frac{44}{1\,105}$	$\frac{1}{8}$
8	$\frac{539}{1\,695}$	$\frac{196}{565}$	$\frac{49}{1\,695}$	$\frac{1}{11}$
10	$\frac{133\,392}{414\,125}$	$\frac{28\,584}{82\,825}$	$\frac{9\,528}{414\,125}$	$\frac{1}{14}$
12	$\frac{6\,443}{19\,675}$	$\frac{6\,822}{19\,675}$	$\frac{379}{19\,675}$	$\frac{1}{17}$
14	$\frac{85\,315\,584}{256\,406\,305}$	$\frac{447\,906\,816}{1\,282\,031\,525}$	$\frac{21\,328\,896}{1\,282\,031\,525}$	$\frac{1}{20}$

The programs and data used to produce Tables 1 and 2 accompany the arXiv version as ancillary files. Instructions are included for reproducing the computations for every even N from 2 through 14.

9 Concluding Remarks

The local Whitehead chain is the random walk on the directed graph of graphs in which every move is counted with multiplicity. Its stationary measure is given by (3.4). The spectral and conductance estimates show that both relaxation and mixing require at least a linear number of steps.

The bridge cut has the exact conductance (7.5). It remains to determine whether there is an absolute $c > 0$ such that

$$\lambda_2(I - P_N) \geq \frac{c}{N}.$$

Together with Theorem 6.2, such a bound would determine the order of the relaxation time.

Finally, one may ask whether the Charlier upper bounds are asymptotically sharp for the individual eigenvalues.

Declaration of interests

The author has no competing interests to declare.

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